

BOUDROUSS RÉDA | CV

Software Engineer – Static Analysis, Compilers & Formal Methods – Paris

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Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, specialized in **static analysis by abstract interpretation**, **formal program verification** and the implementation of programming languages and runtimes. I ported **MOPSA**, an abstract-interpretation analyzer for C and Python developed at LIP6 (OCaml/C/C++), to WebAssembly, and built an abstract-interpretation-based analyzer in **Rust**. At Hexaglobe I industrialized LLM-agent test production gated by **deterministic verification signals** such as mandatory green tests.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python (OCaml/C/C++, developed at LIP6), to **WebAssembly**, compiling the OCaml runtime and the full native stack (Clang frontend, **Apron**, GMP, MPFR) into a single static Wasm module with Emscripten.
- Engineered a two-stage LLVM/Clang cross-compilation (native TableGen tools first, then a Wasm frontend-only build) to embed a C parser in the browser, and generated a static table of ~1,400 C FFI primitives to replace dynamic loading.
- Addressed the **soundness** hazard of Wasm's fixed rounding mode (no fesetround) by recompiling Apron's numerical domains to use exact GMP rationals.
- Shipped a fully client-side [playground](#) and a technical [write-up](#).

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Designed and implemented an **abstract-interpretation**-based static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.
- Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a dedicated CFG-based IR and flags an infinite render loop when the state fixpoint **widens**. Resolves and inlines custom hooks across files, with concrete semantics based on the React-tRace paper (OOPSLA 2025).

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: **static analysis by abstract interpretation** and type systems (course led by Antoine Miné), inference of numerical properties, program specification & **formal verification**, compiler construction, design & implementation of programming languages and runtimes, concurrent & distributed programming.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases and functional programming.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming)

2025 – 2026

- Industrialized **LLM-agent test production** gated by deterministic **verification signals** (mandatory green tests), with sandboxed planner/implementer/reviewer orchestration in Docker, versioned guardrails and systematic human review, producing ~720 integration tests in 3 days.
- Designed objective, automated **verification oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder.
- Brought zero-coverage legacy products to ~**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS), and built **GitLab CI/CD** test pipelines with shared components consumed by 4+ repositories.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.